FUN-3.A.4 Specific rules can be used to find the derivatives for sine, cosine, exponential, and logarithmic functions.
LIM-3.A.1 In some cases, recognizing an expression for the definition of the derivative of a function whose derivative is known offers a strategy for determining a limit.

2-7 Derivatives of Important Functions

Previously, we learned the power rule for derivatives. This rule can be applied if:

1. x is the base
2. power is a constant

However, we will work with many different types of functions where the power rule is not applicable. So, we will spend the next few weeks learning the derivative rules for several types of functions as well as derivative techniques and strategies.

Derivatives of Important Functions	
Function	**Derivative**
$f(x) = \sin(x)$	$f'(x) = \cos x$
$f(x) = \cos(x)$	$f'(x) = -\sin x$
$f(x) = e^x$	$f'(x) = e^x$
$f(x) = \ln(x)$	$f'(x) = \frac{1}{x}$

Example 1: Given the following functions, find $f'(x)$.

a. $f(x) = 2\sin(x)$

$f'(x) = 2\cos x$

b. $f(x) = \ln(x) - \cos(x)$

$f'(x) = \frac{1}{x} + \sin x$

c. $f(x) = e^x - x^2 + 7$

$f'(x) = e^x - 2x$

d. $f(x) = 1 - \sin(x) + \frac{\ln(x)}{3}$

$f'(x) = -\cos x + \frac{1}{3x}$

$\frac{1}{3} \cdot \ln x$

Example 1: Given the following functions, find $f'(x)$.

e. $f(x) = (2x - 7)(3x + 2)$

$= 6x^2 - 17x - 14$

$f'(x) = 12x - 17$

f. $f(x) = \frac{1}{x} + \frac{x}{2} + \frac{\cos(x)}{3} - e^4$

$\frac{x}{2} = \frac{1}{2}x$

$\frac{\cos x}{3} = \frac{1}{3}\cos x$

$f'(x) = \frac{-1}{x^2} + \frac{1}{2} - \frac{\sin x}{3}$

Example 1: Given the following functions, find $f'(x)$.

g. $f(x) = -3e^x + 4\ln(x)$

$f'(x) = -3e^x + \frac{4}{x}$

h. $f(x) = 3\sqrt{x} - \frac{4}{5x^3} - \frac{2}{\sqrt{x}}$ $= 3x^{1/2} - \frac{4}{5}x^{-3} - 2x^{-1/2}$

$f'(x) = \frac{3x^{-1/2}}{2} + \frac{12x^{-4}}{5} + x^{-3/2}$

$= \frac{3x^{-1/2}}{2} + \frac{12}{5x^4} + x^{-3/2}$

i. $f(x) = 4xe^{\ln(1)} - x^{-3}$

$= 4x \cdot e^0 - x^{-3}$

$= 4x - x^{-3}$

$f'(x) = 4 + 3x^{-4}$

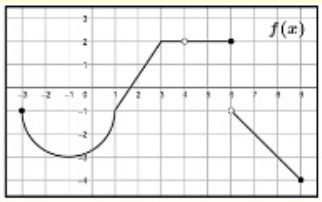
$\ln 1 = 0$

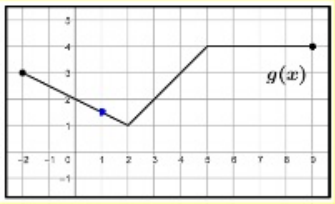
$e^{\ln 5} = 5$

$e^{\ln 3.2} = 3.2$

Example 2: The graphs of the functions $f(x)$ and $g(x)$ are given above along with selected values of the function $h(x)$ and its derivative $h'(x)$. Use the information above to find the following values.

x	-2	-1	0	1	2	4	5	6	8
$h(x)$	4	-1	3	7	-5	2	0	1	-2
$h'(x)$	3	2	-1	4	-3	1	-6	5	2
$h''(x)$	0	-4	3	1	2	7	9	-5	-1





a. $k(x) = \sin(x) - 3g(x) + x^2$ Find $k'(1)$

$k'(x) = \cos x - 3g'(x) + 2x$

$k'(1) = \cos 1 - 3g'(1) + 2$

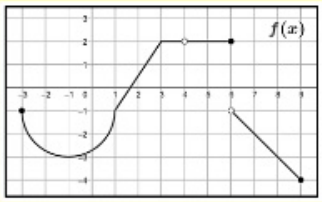
$= \cos 1 - 3\left(-\frac{1}{2}\right) + 2$

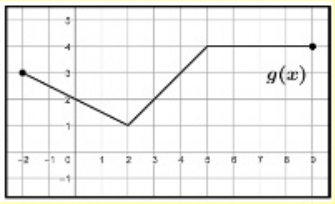
$= \cos 1 + \frac{7}{2}$

$\frac{3}{2} + 2 = \frac{3}{2} + \frac{4}{2}$

Example 2: The graphs of the functions $f(x)$ and $g(x)$ are given above along with selected values of the function $h(x)$ and its derivative $h'(x)$. Use the information above to find the following values.

x	-2	-1	0	1	2	4	5	6	8
$h(x)$	4	-1	3	7	-5	2	0	1	-2
$h'(x)$	3	2	-1	4	-3	1	-6	5	2
$h''(x)$	0	-4	3	1	2	7	9	-5	-1





b. $w(x) = -5\cos(x) + \dfrac{3}{\sqrt{x}}$ Find $w'(4)$

$3x^{-1/2}$

$w'(x) = 5\sin x - \dfrac{3x^{-3/2}}{2}$

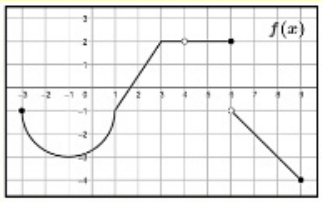
$4^{-3/2} = \dfrac{1}{4^{3/2}} = \dfrac{1}{(\sqrt{4})^3} = \dfrac{1}{8}$

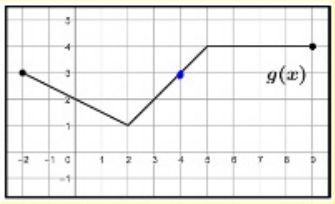
$w'(4) = 5\sin 4 - \dfrac{3(4)^{-3/2}}{2}$ $\longleftarrow \dfrac{3}{2}\cdot\dfrac{1}{8}$

$= 5\sin 4 - \dfrac{3}{16}$

Example 2: The graphs of the functions $f(x)$ and $g(x)$ are given above along with selected values of the function $h(x)$ and its derivative $h'(x)$. Use the information above to find the following values.

x	-2	-1	0	1	2	4	5	6	8
$h(x)$	4	-1	3	7	-5	2	0	1	-2
$h'(x)$	3	2	-1	4	-3	1	-6	5	2
$h''(x)$	0	-4	3	1	2	7	9	-5	-1





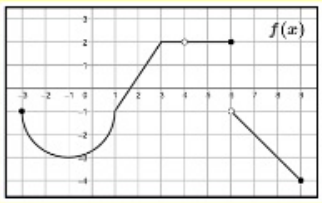
c. $n(x) = h(x) - g(x) + 1$ Find $n'(4)$

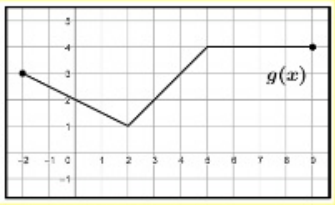
$n'(x) = h'(x) - g'(x)$

$n'(4) = h'(4) - g'(4)$

$= 1 - 1 = \boxed{0}$

Example 2: The graphs of the functions $f(x)$ and $g(x)$ are given above along with selected values of the function $h(x)$ and its derivative $h'(x)$. Use the information above to find the following values.





x	-2	-1	0	1	2	4	5	6	8
$h(x)$	4	-1	3	7	-5	2	0	1	-2
$h'(x)$	3	2	-1	4	-3	1	-6	5	2
$h''(x)$	0	-4	3	1	2	7	9	-5	-1

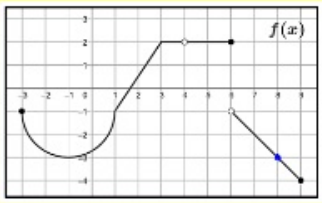
d. $p(x) = 7 - e^x + 2h'(x)$ Find $p'(0)$

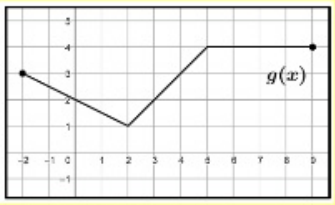
$p'(x) = -e^x + 2h''(x)$

$p'(0) = -e^0 + 2h''(0)$

$= -1 + 2(3) = -1 + 6 = \boxed{5}$

Example 2: The graphs of the functions $f(x)$ and $g(x)$ are given above along with selected values of the function $h(x)$ and its derivative $h'(x)$. Use the information above to find the following values.





x	-2	-1	0	1	2	4	5	6	8
$h(x)$	4	-1	3	7	-5	2	0	1	-2
$h'(x)$	3	2	-1	4	-3	1	-6	5	2
$h''(x)$	0	-4	3	1	2	7	9	-5	-1

e. $q(x) = f(x) + h(1) - \ln(x)$ Find $q'(8)$

$= f(x) + 7 - \ln(x)$

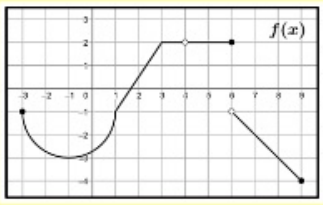
$q'(x) = f'(x) - \frac{1}{x}$

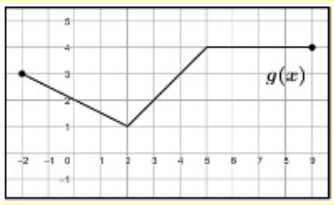
$q'(8) = f'(8) - \frac{1}{8}$

$= -1 - \frac{1}{8} = -\frac{8}{8} - \frac{1}{8} = \frac{-9}{8}$

Example 2: The graphs of the functions $f(x)$ and $g(x)$ are given above along with selected values of the function $h(x)$ and its derivative $h'(x)$. Use the information above to find the following values.

x	-2	-1	0	1	2	4	5	6	8
$h(x)$	4	-1	3	7	-5	2	0	1	-2
$h'(x)$	3	2	-1	4	-3	1	-6	5	2
$h''(x)$	0	-4	3	1	2	7	9	-5	-1





f. $r(x) = f(x) + 2\cos(x) - \dfrac{1}{x^2}$ Find $r'(3)$

$-1x^{-2}$

$r'(x) = f'(x) - 2\sin x + 2x^{-3}$

$x^{-3} = \frac{1}{x^3}$

$r'(3) = \underbrace{f'(3)}_{\text{undefined}} - 2\sin 3 + \frac{2}{27}$

$r'(3) = \text{undefined}$